Comparative Analysis of Spatial Transcriptomics Workflows

GitHub Link: https://github.com/Rameensajjad/SPATIAL-TRANSCRIPTOMICS

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Introduction

Spatial transcriptomics has rapidly diversified into a family of technologies that differ fundamentally in spatial resolution, molecular panel breadth, imaging modality, and dataset scale. This report provides a structured comparative analysis of four published analytical workflows spanning this technological spectrum from the foundational Scanpy-based Visium pipeline through progressively more sophisticated Squidpy-based analyses, culminating in a large-scale Xenium in situ workflow built on the modern SpatialData framework. The comparison is organized across seven analytical dimensions: data architecture, quality control, preprocessing and clustering, image analysis, spatial graph statistics, biological findings, and computational considerations. The central conclusion is that each successive workflow adds a distinct, non-redundant analytical layer atop a shared Scanpy preprocessing backbone, and that the choice of analytical framework must be matched to the resolution, scale, and imaging modality of the underlying technology.

1. Scope and Workflow Overview

The four workflows reviewed are:

Scanpy Visium Analysis — Human Lymph Node introduces the foundational spatial transcriptomics pipeline using 10x Genomics Visium data from a human lymph node tissue section. The analysis is implemented entirely in Scanpy and AnnData, covering quality control, normalization, dimensionality reduction, clustering, spatial visualization, and marker gene identification. It serves as the entry point to the field and the shared methodological baseline for all subsequent workflows.

Squidpy Visium Fluorescence Analysis — Mouse Brain extends the baseline pipeline to a fluorescence-imaged Visium dataset of a mouse brain coronal section. The primary analytical contribution is the integration of quantitative tissue image analysis into the spatial transcriptomics workflow — including nucleus segmentation, cell counting, fluorescence channel intensity quantification, and multi-scale image feature extraction — demonstrating that the tissue image itself contains biologically informative signal complementary to gene expression.

Squidpy Visium H&E Analysis — Mouse Brain applies the full Squidpy spatial graph analysis toolkit to the same mouse brain system using a conventional H&E-stained Visium dataset. This workflow introduces neighborhood enrichment, co-occurrence analysis, ligand-receptor interaction inference, and spatially variable gene detection via Moran's I collectively transforming the analysis from descriptive spatial visualization into quantitative spatial statistics.

Squidpy Xenium In Situ Analysis — Human Lung Cancer addresses the challenge of single-cell resolved, large-scale spatial transcriptomics data using 10x Genomics Xenium data from FFPE human lung cancer tissue. The workflow adopts the SpatialData framework and Zarr-backed on-disk storage to handle 161,000 single cells and 40 million individual transcript detections, introduces Delaunay triangulation-based spatial graph construction, and extends the analytical toolkit with centrality scores and interactive visualization via napari-spatialdata.

2. Data Architecture and Loading

The four workflows reveal a clear progression in the complexity of data structures required to represent increasingly information-rich spatial datasets.

The Scanpy Visium workflow uses the simplest possible structure: a single **AnnData** object loaded through a one-line built-in function. The AnnData container stores the gene expression count matrix, per-spot metadata, the H&E tissue image, and spatial coordinates in a single unified object. For introductory spatial transcriptomics, this structure is entirely sufficient and imposes minimal cognitive overhead.

The two Squidpy Visium workflows introduce a second object alongside AnnData: Squidpy's **ImageContainer**, which holds the tissue image in a dedicated structure designed for efficient cropping, multi-layer processing, and channel-wise operations. This separation is deliberate the ImageContainer exposes image processing capabilities that would be cumbersome to implement if the image were embedded within AnnData's unstructured metadata slot. The two-object paradigm cleanly divides responsibilities: one object owns transcriptomics data and cell metadata; the other owns spatial imaging data and derived image layers.

The Xenium workflow necessitates the most sophisticated data architecture: the **SpatialData** framework backed by on-disk **Zarr** storage. The scale of Xenium data — 161,000 cells, 40 million individual transcript point detections, and a full-resolution tissue image spanning over one billion pixels stored across five resolution levels, renders conventional in-memory AnnData structures impractical. The SpatialData object unifies five distinct data modalities within a single coordinate-aligned framework:

- **Images:** A multi-scale morphology focus image at five resolution levels, enabling efficient rendering at different zoom levels without full-resolution loading
- **Labels:** Pre-computed cell and nucleus segmentation masks, where each pixel is assigned the integer identifier of the cell it belongs to
- **Points:** A DataFrame of 40,257,199 individual transcript detections, each with 3D spatial coordinates and gene identity
- **Shapes:** Cell boundary polygons, nucleus boundary polygons, and simplified cell circle approximations for 161,000 cells
- **Tables:** An AnnData object containing the per-cell gene expression count matrix (161,000 × 480) and per-cell metadata

All data modalities are registered to a shared global coordinate system, enabling spatially coherent joint analysis across all layers. Downstream operations stream from the Zarr store rather than loading everything into memory, which is what makes this scale tractable.

A secondary architectural note concerns the MERFISH secondary analysis embedded within the Scanpy Visium workflow. This demonstrates the opposite end of the complexity spectrum, manually constructing an AnnData object from two separate CSV and Excel files, transposing the count matrix, aligning indices, and manually assigning spatial coordinates. This manual assembly highlights, by contrast, how much data engineering the platform-specific readers in spatialdata-io automate for Xenium data.

3. Dataset Scale and Technological Resolution

The profiled datasets differ dramatically in scale and spatial resolution, and these differences directly determine which analyses are feasible and which require computational adaptation.

The Visium platform operates at **spot resolution**, where each measurement represents a 55 μm diameter tissue area that may capture RNA from multiple cells simultaneously. The human lymph node Visium dataset contains 4,035 spots across 36,601 genes. The fluorescence mouse brain workflow uses a cropped subset of 704 spots from the same platform and panel scale, while the H&E mouse brain workflow contains 2,688 spots. All three Visium datasets are compact enough to be managed entirely in memory and support whole-transcriptome profiling at the cost of single-cell resolution.

The Xenium platform operates at **single-cell resolution**, detecting individual RNA molecules and assigning them to segmented cell boundaries produced by the instrument's onboard image analysis pipeline. The 161,000 cells profiled across the lung cancer tissue section represent a 40-fold increase in observation count over the lymph node Visium dataset. However, the gene panel is deliberately narrow, 480 targeted genes versus tens of thousands, reflecting the fundamental trade-off between spatial and cellular resolution on the one hand and molecular panel breadth on the other that characterizes the current landscape of spatial transcriptomics technologies.

The MERFISH dataset, used as a secondary example within the Scanpy workflow, represents an intermediate point: 645 single cells profiled across 12,903 genes, higher cellular resolution than Visium but a far smaller absolute scale than Xenium, appropriate for the single-cell-type (U2-OS) culture experiment from which it was derived.

4. Quality Control

Quality control strategies are broadly shared across all workflows but differ in the specific metrics available and the thresholds applied, reflecting differences in underlying biology, technology, and the degree of data pre-processing performed upstream.

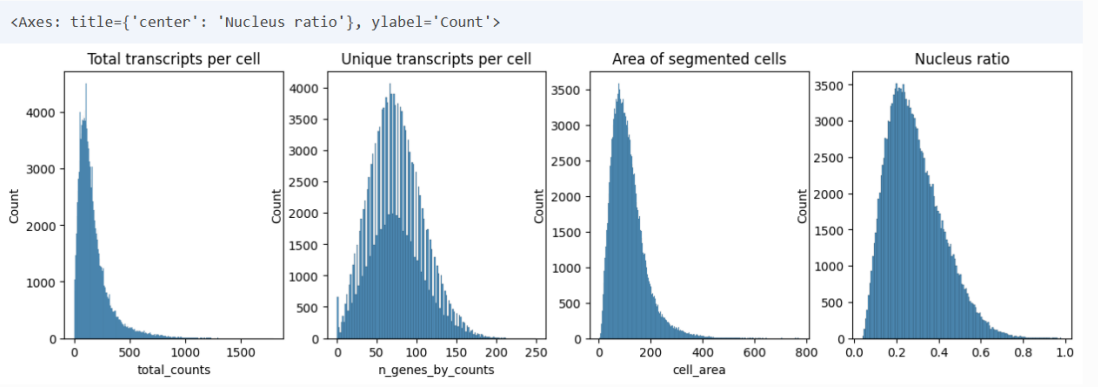
All four workflows compute total counts per observation and the number of detected genes or unique features per observation, and apply minimum count thresholds to remove low-quality cells or spots. Four-panel histogram figures a consistent visual pattern across all workflows guide threshold selection before filtering is applied.

The Scanpy Visium workflow introduces **mitochondrial read percentage** as a critical QC filter, flagging genes whose names begin with "MT-" and removing spots where more than 20% of total counts derive from mitochondrial transcripts. This metric is standard in single-cell RNA sequencing and directly applicable to whole-transcriptome Visium data, where mitochondrial genes are profiled alongside nuclear genes. Elevated mitochondrial proportions indicate damaged, necrotic, or poorly captured tissue regions. This filter is not applied in the Xenium workflow, where targeted gene panels typically exclude mitochondrial genes by design.

The Xenium workflow introduces platform-specific quality control metrics not applicable to any Visium analysis: **negative control probe counts** (DNA probes designed to bind nothing in the tissue, providing an estimate of non-specific hybridization background), **negative decoding counts** (errors in the combinatorial fluorescence decoding process), and **unassigned and deprecated codeword counts**. Computing the fraction of total counts attributable to these control elements 0.005% for control probes and 0.003% for control codewords in this dataset provides a dataset-level signal-to-noise assessment before per-cell filtering begins. These extremely low values confirm high platform fidelity.

Additionally, because the Xenium instrument produces precise cell and nucleus boundary polygons for every detected cell, two morphological quality metrics become available: **cell area** (the physical size of each segmented cell) and the **nucleus-to-cell area ratio** (the fraction of total cell area occupied by the nucleus). Very small cells may represent over-segmented fragments; very large cells may be under-segmented doublets. Anomalous nucleus ratios may indicate bare nuclei captured without cytoplasm, or cells where the nucleus was not properly delineated. These morphological filters add a dimension of quality assessment entirely absent from spot-based Visium workflows.

Filtering thresholds reflect the different gene panel scales across technologies. The Scanpy Visium workflow applies stringent count boundaries of 5,000–35,000 total counts per spot, appropriate for whole-transcriptome profiling. The Xenium workflow applies a permissive minimum of just 10 counts, appropriate for a 480-gene targeted panel where well-captured cells may have far fewer total transcripts than a whole-transcriptome Visium spot. The fluorescence and H&E Visium workflows use pre-processed, pre-filtered datasets, meaning that QC decisions were made upstream and the workflows begin from already-clean data.



5. Preprocessing, Dimensionality Reduction, and Clustering

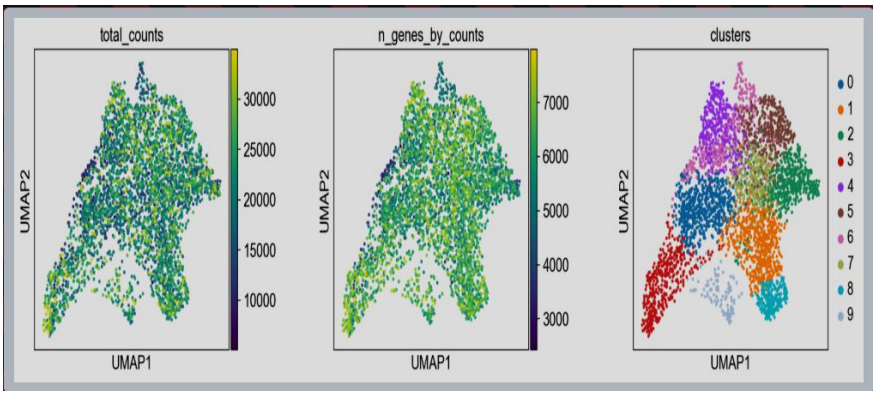
The core preprocessing pipeline is effectively identical across all four workflows, reflecting the maturity and broad adoption of the Scanpy framework for transcriptomics data regardless of the spatial platform from which it originated.

All workflows apply **total-count normalization** to place observations on a comparable expression scale, followed by a **log-plus-one transformation** to stabilize variance and compress the dynamic range of expression values. **Principal Component Analysis (PCA)** then reduces the high-dimensional gene expression matrix to a tractable set of principal components capturing the dominant axes of transcriptional variation. A **neighbor graph** is constructed from the PCA coordinates, connecting observations that are transcriptionally similar. **UMAP** produces a 2D embedding of this graph for visualization, and **Leiden clustering** partitions the graph into transcriptionally distinct communities. This shared vocabulary is a deliberate design choice of the ecosystem the same pipeline applies from a 645-cell MERFISH experiment to a 161,000-cell Xenium dataset.

Meaningful differences exist in parameter choices. The Scanpy Visium and Xenium workflows both use 50 PCA components as the default, though the information content of these components differs substantially: 50 components from 36,601 whole-transcriptome genes captures a different proportion of variance than 50 components from 480 targeted genes. The MERFISH secondary analysis uses only 15 components, appropriate for a small, low-complexity dataset. The image feature clustering performed in the fluorescence Visium workflow uses 10 components, reflecting the low intrinsic dimensionality of extracted image statistics.

A critical distinction in the Xenium workflow is the explicit preservation of **raw integer counts** in a separate AnnData layer before normalization. The operation copies the raw count matrix to adata.layers["counts"] before any transformation is applied. This is essential because certain downstream methods including some differential expression models and trajectory inference algorithms require access to the original integer-valued counts and cannot operate correctly on normalized or log-transformed data. This step is absent from the Visium workflows, reflecting the more stringent data provenance requirements of large-scale in situ analysis pipelines.

UMAP visualizations serve a consistent validation function across all workflows: plotting the embedding colored by total counts, detected gene numbers, and cluster identity simultaneously allows the analyst to verify that cluster boundaries are not driven by technical gradients in sequencing depth. The absence of strong technical overlaps with cluster structure is a prerequisite for confident biological interpretation in all four cases.



6. Imaging and Image Analysis

The treatment of tissue images represents one of the most substantial analytical differentiators across the four workflows, ranging from a purely passive visualization role to a primary source of quantitative biological inference.

6.1 H&E as a Visualization Backdrop

In the Scanpy Visium workflow, the H&E tissue image functions exclusively as a **passive backdrop for spatial visualization**. The image is stored within the AnnData object and rendered beneath spot-level overlays of cluster labels, gene expression values, and QC metrics. No pixel-level computation is performed on the image; it contributes context for interpreting where clusters are anatomically located but plays no role in the analytical pipeline itself. This approach is appropriate as an introduction but leaves substantial information in the image unexploited.

6.2 Fluorescence Image as Primary Analytical Input

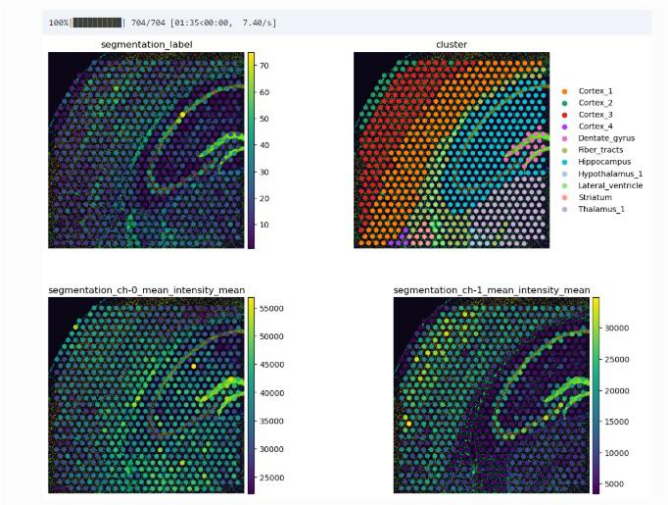
The fluorescence Visium workflow elevates the tissue image from background to **primary analytical input**. The three-channel fluorescence image DAPI (all cell nuclei), anti-NEUN (neurons), and anti-GFAP (glial cells) is processed through a complete image analysis pipeline:

Image smoothing via Gaussian filtering reduces high-frequency pixel noise before segmentation, preventing the watershed algorithm from over-fragmenting nuclei or generating spurious detections.

Watershed segmentation is applied to the DAPI channel of the smoothed image. The watershed algorithm treats pixel intensity as a topographical surface, identifying individual nuclei as basins separated by ridges of lower intensity. The result is a label image where each pixel belonging to a distinct segmented nucleus receives a unique integer identifier. The quality of segmentation is validated visually by comparing the raw DAPI channel against the label image in a 500×500 pixel crop.

Segmentation feature extraction computes, for each Visium spot, the number of segmented nuclei (a direct proxy for cell count) and the mean fluorescence intensity of each channel within

segmented nuclei — providing per-spot estimates of neuronal density (NEUN channel) and glial density (GFAP channel).

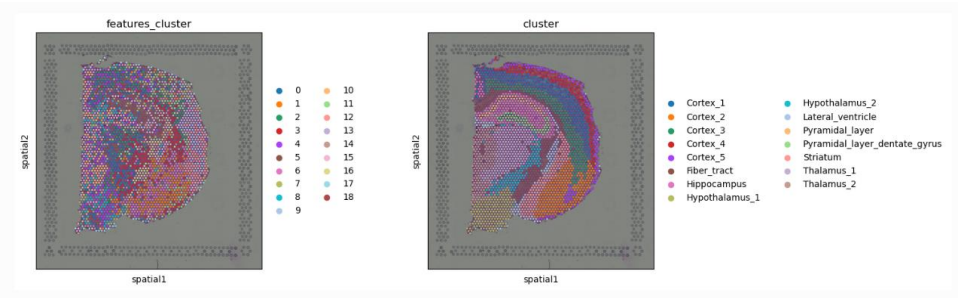


Multi-scale image feature extraction computes summary statistics, pixel intensity histograms, and gray-level co-occurrence matrix texture descriptors at two spatial scales: one capturing the tissue directly beneath each spot at full resolution (using a circular mask), and one incorporating a larger surrounding tissue context at reduced resolution. This multi-scale approach captures morphological information simultaneously at local and regional granularities.

The resulting image features are combined into a single per-spot feature matrix, scaled, reduced via PCA, and subjected to Leiden clustering producing image-based cluster assignments independent of gene expression.

6.3 H&E Feature Extraction for Comparative Clustering

The H&E Visium workflow applies a subset of image feature extraction — summary statistics at two scales — to generate an image-based Leiden clustering that is compared directly against the gene expression clusters. Because H&E staining is not molecularly specific, the features encode morphological structure (cell density, nuclear size and staining intensity, tissue texture) rather than cell-type identity. The comparison reveals that image features broadly recapitulate gene expression clusters in anatomically distinct regions (fiber tracts, hippocampal boundaries) while diverging in the cortex, where gene expression resolves laminar cortical layers that appear morphologically similar in H&E images. This divergence is itself biologically informative.



6.4 Pre-Computed Segmentation in Xenium

The Xenium workflow includes a high-resolution morphology image stored across five resolution levels in the SpatialData object, but the workflow does not perform analytical image feature extraction from it. Instead, cell and nucleus segmentation is provided as pre-computed label images produced by the Xenium instrument's onboard pipeline. This eliminates the need for post-hoc segmentation that is central to the fluorescence Visium workflow. The morphology image serves as a visual reference in spatialdata-plot visualizations, where gene expression is overlaid directly on the tissue morphology but the image is not processed as an analytical input in the manner of the fluorescence workflow.

7. Spatial Graph Construction and Statistical Analysis

The most analytically sophisticated dimension across the four workflows is the use of spatial neighbor graphs and graph-based statistical methods. This is where the Squidpy-based workflows diverge most substantively from the Scanpy baseline.

7.1 Absence of Graph Analysis in the Baseline Workflow

The Scanpy Visium workflow constructs no spatial graph. Spatial context is conveyed entirely through visualization overlaying analytical results on the tissue image and no quantitative spatial statistics are computed. This is appropriate for an introductory workflow but represents a significant underutilization of the spatial coordinate information available in Visium data.

7.2 Spatial Graph Construction: Grid-Based versus Delaunay

Both Squidpy Visium workflows construct spatial neighbor graphs using squidpy.gr.spatial_neighbors() with default parameters appropriate for the regular hexagonal array of Visium spots, where grid-based neighborhood definition is geometrically natural.

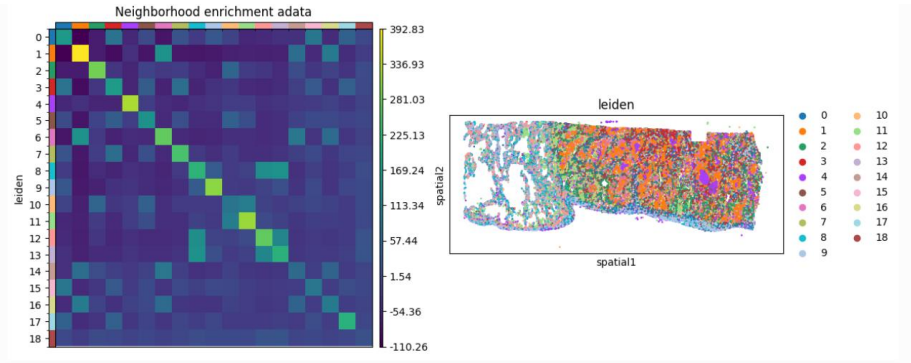
The Xenium workflow requires a methodologically distinct approach: **Delaunay triangulation**, specified via coord_type="generic" and delaunay=True. Delaunay triangulation connects each cell to its geometrically natural nearest neighbors by finding the triangulation that maximizes the minimum angle of all triangles formed by cell centroids. This approach is appropriate for irregular single-cell coordinates produced by image segmentation, where cells are distributed continuously across the tissue rather than placed on a fixed array. Using a grid-based graph for Xenium data would be methodologically incorrect; using Delaunay triangulation for Visium data would ignore the structured geometry of the spot array. The parameter choice is therefore a critical adaptation not a stylistic preference to the underlying spatial structure of the data.

7.3 Neighborhood Enrichment Analysis

Neighborhood enrichment analysis quantifies whether spots or cells from two different clusters are spatially adjacent more or less frequently than expected under a random arrangement of cluster labels. The enrichment score is computed via a permutation test: cluster labels are shuffled across observations a specified number of times (1,000 permutations by default), and the observed co-

adjacency frequency between each cluster pair is compared against this null distribution to generate a z-score. The result is visualized as a symmetric matrix heatmap.

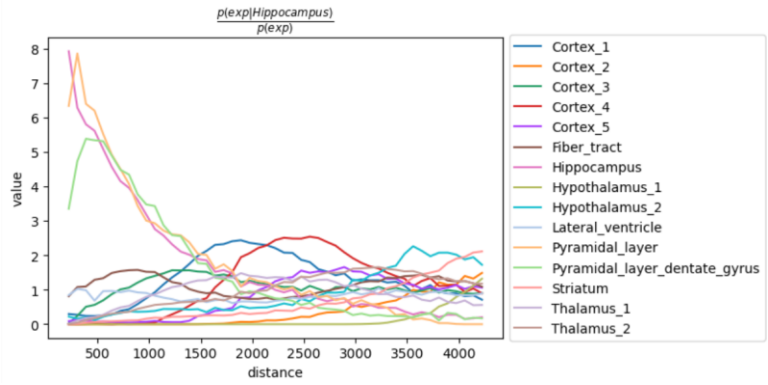
This method appears in both the H&E Visium and Xenium workflows. In the mouse brain, it recovers the expected neuroanatomical relationship: the Pyramidal_layer_dentate_gyrus and Pyramidal_layer clusters are frequently adjacent to the broader Hippocampus cluster, consistent with their known embedding within the hippocampal formation. In the lung cancer tissue, the same method characterizes the spatial organization of Leiden clusters within the tumor microenvironment, where the biological interpretation requires domain knowledge of tumor architecture rather than classical anatomy.



7.4 Co-occurrence Analysis

Co-occurrence analysis evaluates how the probability of observing one cluster changes as a function of distance from another cluster, operating directly on continuous spatial coordinates rather than the discrete neighbor graph. The score at any radius is the ratio of the conditional probability of observing cluster *exp* near cluster *cond* to the unconditional probability of observing cluster *exp* at that scale producing a distance-dependent curve where values above 1 indicate spatial association and values below 1 indicate spatial exclusion.

This analysis appears in both the H&E Visium and Xenium workflows. A critical computational adaptation is introduced in the Xenium workflow: because computing pairwise distances among 161,000 cells would require constructing a 161,000 × 161,000 distance matrix (exceeding available memory), a 50% random subsample is created and stored as a new table in the SpatialData object before the analysis is run. Squidpy further automatically decomposes the distance computation into 39 sequential sub-batches to prevent memory exhaustion. No such adaptation is required for the Visium datasets, where the spot counts are three orders of magnitude smaller.



7.5 Ligand-Receptor Interaction Analysis

The H&E Visium workflow is the only analysis to deploy **ligand-receptor interaction analysis**, using Squidpy's reimplementation of CellPhoneDB extended with the Omnipath signaling database. The method tests, for each pair of clusters and each annotated ligand-receptor pair in the database, whether the mean expression of the ligand in one cluster and the receptor in the other exceeds what is expected under random cluster label permutation. The output is visualized as a dot plot filtered to high-expression, statistically significant interactions, focused on the Hippocampus cluster as the source and its two pyramidal sub-layer clusters as targets.

This analysis extends the spatial proximity findings established by neighborhood enrichment and co-occurrence into the molecular domain, identifying candidate intercellular signaling events that may drive or reflect the observed spatial adjacency. Its absence from the Xenium workflow reflects the tutorial's different analytical priorities rather than any methodological incompatibility, as ligand-receptor analysis on Xenium data is entirely feasible.

7.6 Spatially Variable Gene Detection via Moran's I

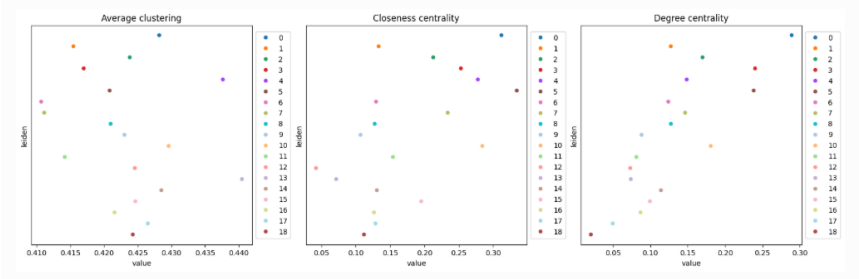
Moran's I is a classical spatial autocorrelation statistic that quantifies whether a gene's expression across observations follows a non-random spatial pattern. A value near +1 indicates strong positive spatial autocorrelation observations with high expression of that gene cluster together in space. A value near 0 indicates random spatial distribution. Genes with high Moran's I and statistically significant permutation-based p-values are classified as spatially variable.

Both the H&E Visium and Xenium workflows apply squidpy.gr.spatial_autocorr() in Moran's I mode with 100 permutations, but to very different gene panels and biological contexts. In the mouse brain, 1,000 highly variable genes from the whole-transcriptome panel are evaluated; top-ranked spatially variable genes include Plp1 (a myelin gene enriched in oligodendrocytes along white matter fiber tracts), Olfm1, Snap25 (a synaptic gene enriched in neurons), and Itpka all with Moran's I values above 0.67. In the lung cancer tissue, all 480 targeted genes are evaluated; top-ranked genes include AREG and MET (epithelial growth factor pathway genes), EPCAM (canonical epithelial marker), IGKC and IGHG1 (immunoglobulin subunit genes marking B cells and plasma cells), and IDO1 (an immunosuppressive enzyme associated with tumor immune evasion)

collectively reflecting the known cellular heterogeneity and compartmentalization of the lung cancer tumor microenvironment.

7.7 Centrality Scores

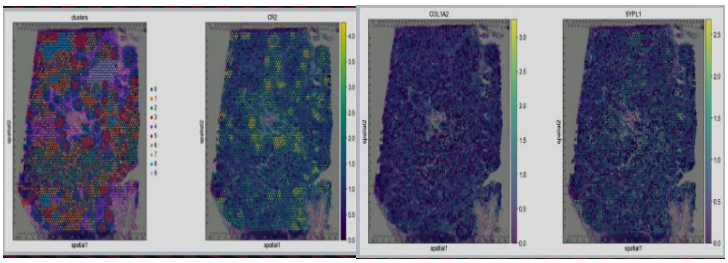
Centrality scores are exclusive to the Xenium workflow, representing the only method across the four analyses that characterizes how individual cluster populations are positioned within the global spatial graph. Three scores are computed: **closeness centrality** (how close, on average, a cluster's cells are to all other cells in the tissue graph reflecting centrality versus peripherality within the tissue architecture), **degree centrality** (the fraction of cells from other clusters that are directly connected to cells in the focal cluster reflecting how much a cluster interfaces with other populations), and the **clustering coefficient** (the degree to which a cluster's cells form dense, compact spatial groupings versus dispersed distributions). These metrics are most informative at the single-cell scale of Xenium, where tens of thousands of cells populate the graph, and less meaningful at the spot scale of Visium, where graph nodes represent multi-cellular tissue areas.



8. Biological Findings

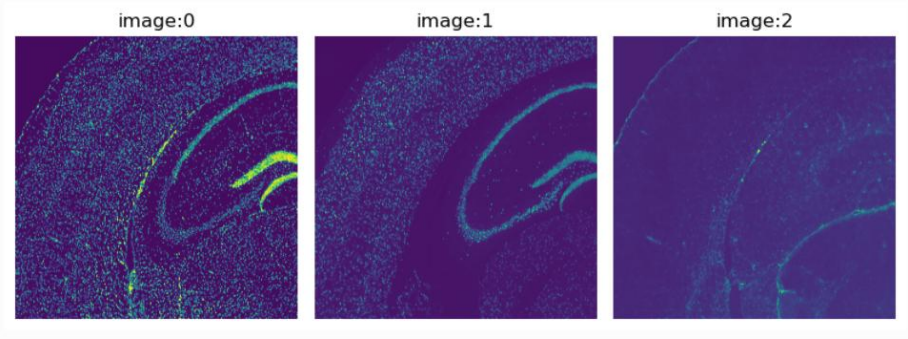
8.1 Human Lymph Node — Transcriptional Clusters Align with Tissue Anatomy

The Scanpy Visium analysis of the human lymph node establishes the foundational finding of spatial transcriptomics: transcriptionally distinct clusters, identified purely from gene expression data, occupy spatially coherent regions of the tissue. The marker gene CR2 is identified as the top differentially expressed gene for cluster 9, and its expression pattern, when mapped spatially, closely mirrors the spatial distribution of that cluster across the tissue section providing a direct validation that transcriptional identity has an anatomical footprint in the lymph node. Spatial relationships between clusters including the consistent adjacency of cluster 5 and cluster 0 are observed visually, suggesting potential cellular interactions in the lymphoid tissue microenvironment.



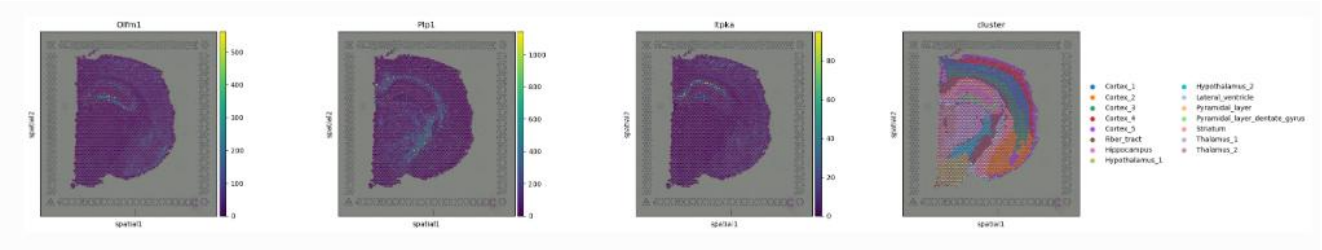
8.2 Mouse Brain (Fluorescence) — Image-Derived Insights Complement and Extend Gene Expression

The fluorescence Visium analysis recovers biological insights inaccessible from gene expression data alone. The per-spot cell count map derived from DAPI nucleus segmentation reveals the dense pyramidal cell layer of the Hippocampus at higher spatial resolution than the gene expression clustering provides, where the entire hippocampal region is assigned to a single cluster. Channel-specific intensity maps confirm the expected neurobiological distribution of cell types: cortical regions labeled Cortex_1 and Cortex_3 show elevated NEUN signal, indicating neuronal enrichment, while fiber tract and lateral ventricle regions show elevated GFAP signal, indicating glial enrichment. These findings are consistent with established neurobiology and are recovered exclusively from the fluorescence image, demonstrating that molecularly specific staining provides an orthogonal and complementary data layer.



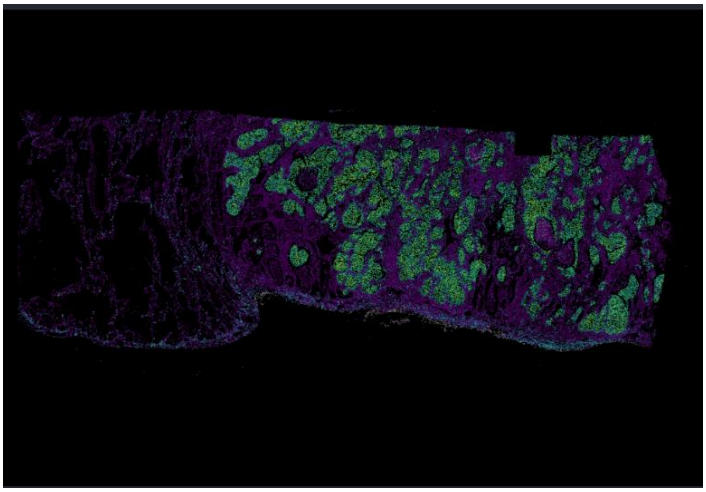
8.3 Mouse Brain (H&E) — Quantitative Spatial Statistics Reveal Intercellular Organization

The H&E Visium analysis produces the richest set of biological insights among the four workflows for a single tissue type. Neighborhood enrichment confirms that the Pyramidal_layer_dentate_gyrus and Pyramidal_layer clusters are frequently adjacent to the broader Hippocampus cluster an anatomically expected finding that serves as a method validation. Co-occurrence analysis quantifies this relationship across spatial scales, showing that Pyramidal_layer co-occurrence with Hippocampus is strongest at short distances and diminishes with increasing radius. Ligand-receptor analysis identifies candidate signaling interactions between the Hippocampus and its pyramidal sub-layers, providing molecular hypotheses about intercellular communication in this brain region. Spatially variable gene detection identifies Plp1 and Snap25 as highly spatially autocorrelated, correctly reflecting the known compartmentalization of myelinating oligodendrocytes in fiber tracts and neurons in cortical and hippocampal layers.



8.4 Human Lung Cancer — Tumor Microenvironment Heterogeneity at Single-Cell Resolution

The Xenium analysis recovers spatially resolved features of the lung cancer tumor microenvironment at single-cell resolution. Spatially variable gene detection identifies a biologically coherent set of top-ranked genes: EPCAM, AREG, and MET reflect the localized distribution of epithelial tumor cell nests; IGKC and IGHG1 mark B cell and plasma cell populations, consistent with the known presence of tertiary lymphoid structures in some lung cancers; IDO1 marks a localized immunosuppressive niche. Centrality score analysis characterizes how Leiden cluster populations are positioned within the tumor spatial graph, distinguishing centrally located cell populations from peripheral ones and identifying clusters with high inter-cluster connectivity potentially marking interface cell populations at tumor-stroma or tumor-immune boundaries.



9. Visualization

All workflows share the core approach of overlaying analytical results on tissue spatial coordinates using either Scanpy's pl.spatial() or Squidpy's pl.spatial_scatter(). The latter is available in the Squidpy-based workflows and provides a superset of the former's capabilities, including multi-panel layouts, finer marker shape control, and compatibility with SpatialData coordinate systems.

Beyond this shared foundation, the Xenium workflow introduces two additional visualization frameworks that have no equivalent in the Visium analyses. **Spatialdata-plot** renders gene expression data as colored cell circles overlaid directly on the full-resolution morphology focus image, enabling direct visual correlation between transcript-level expression and tissue morphology at the level of individual cells. Each gene is rendered in a separate figure panel, and the full tissue extent is visible at the instrument's native image resolution. **Napari-spatialdata** provides a fully interactive graphical interface where all data modalities of the SpatialData object images at multiple resolutions, cell and nucleus segmentation labels, individual transcript point clouds, cell boundary shapes, and expression overlays can be explored interactively with pan, zoom, layer toggling, and real-time expression rendering. This interactive capability is particularly valuable for

hypothesis generation from large datasets where static plots at full tissue scale may obscure local spatial patterns visible only at higher magnification.

10. Computational Considerations

The four workflows make substantially different demands on computational resources, and several explicit strategies for managing large-scale data appear in the Xenium analysis.

The Scanpy Visium workflow is computationally lightweight: all operations complete within seconds to minutes, the entire dataset fits in memory without difficulty, and no parallelization or memory management strategies are required.

The two Squidpy Visium workflows are moderately demanding. Image feature extraction across 704 spots (fluorescence workflow) and 2,688 spots (H&E workflow) requires several minutes per feature set. The ligand-receptor permutation test in the H&E workflow runs in approximately 16 seconds with 100 permutations. Moran's I computation across 1,000 genes with 100 permutations requires approximately 47 seconds.

The Xenium workflow presents the most significant computational challenges and addresses them through four explicit strategies: **on-disk Zarr storage** prevents the full dataset from being loaded into memory; **50% subsampling** before co-occurrence analysis reduces pairwise distance computation from $O(161{,}000^2)$ to $O(80{,}000^2)$; **automatic sub-batch decomposition** of the co-occurrence distance matrix into 39 sequential blocks prevents memory exhaustion; and **parallelization parameters** (n_jobs) are available throughout the Squidpy API but set to 1 in the tutorial for reproducibility. Moran's I computation across all 480 genes with 100 permutations on the subsampled data requires approximately 5 minutes and 49 seconds substantially longer per gene than the Visium equivalent due to the far greater number of observations in the spatial graph.

11. Analytical Progression and Methodological Summary

The following table consolidates the methodological contributions of each workflow:

Analytical Dimension	Scanpy Visium (Lymph Node)	Squidpy Visium Fluo (Brain)	Squidpy Visium H&E (Brain)	Squidpy Xenium (Lung Cancer)
Technology	Visium	Visium Fluorescence	Visium H&E	Xenium In Situ
Resolution	Spot (55 µm)	Spot (55 µm)	Spot (55 µm)	Single cell
Observations	4,035 spots	704 spots (crop)	2,688 spots	161,000 cells

Analytical Dimension	Scanpy Visium (Lymph Node)	Squidpy Visium Fluo (Brain)	Squidpy Visium H&E (Brain)	Squidpy Xenium (Lung Cancer)
Gene panel	36,601 (WTA)	Whole transcriptome	Whole transcriptome	480 targeted
Data framework	AnnData	AnnData + ImageContainer	AnnData + ImageContainer	SpatialData + Zarr
Image analysis	None	Full (segmentation + features)	Partial (summary features)	Pre-computed segmentation
Spatial graph	None	None	Grid-based	Delaunay triangulation
Neighborhood enrichment	No	No	Yes	Yes
Co-occurrence	No	No	Yes	Yes (subsampled)
Ligand-receptor	No	No	Yes	No
Moran's I	No	No	Yes (1,000 genes)	Yes (480 genes)
Centrality scores	No	No	No	Yes
Interactive visualization	No	No	No	Napari
Secondary technology	MERFISH	—	—	—

12. Conclusion

The progression from the Scanpy baseline to the Xenium workflow is not linear in a single dimension each workflow adds a distinct and non-redundant analytical contribution. The Scanpy Visium workflow establishes the universal preprocessing pipeline and the core spatial transcriptomics result that transcriptional clusters have anatomical identities. The fluorescence Visium workflow demonstrates that molecularly specific tissue images contain quantitative biological information orthogonal to gene expression. The H&E Visium workflow introduces the full toolkit of spatial graph statistics, transforming spatial analysis from description to quantitative inference. The Xenium workflow addresses the challenges of single-cell resolution and large-scale

data through modern data architecture, appropriate graph construction methods, and interactive visualization infrastructure.

The shared Scanpy preprocessing backbone normalization, log transformation, PCA, neighbor graph, UMAP, Leiden ensures that methodological skills and analytical code developed at the introductory level transfer directly to the most advanced workflows in this comparative series. This architectural consistency is a defining strength of the Scanpy and Squidpy ecosystem and reflects a deliberate design philosophy: any spatial transcriptomics dataset, regardless of the technology that generated it, can be analyzed within the same conceptual framework, with technology-specific adaptations introduced modularly rather than requiring fundamentally different analytical approaches.